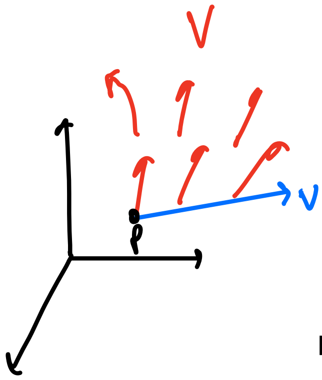
Differential Geometry 5 Sep

Last time:



covariant derivative of a vector field

$$(\nabla_v V)_p = \frac{d}{dt}\Big|_{t=0} V(p+tv).$$

tangent vector at p.

Brutal

A frame field on $\mathbb{R}^3$ is a triple of vector fields E_1, E_2, E_3 which are an orthonormal basis for $T_p\mathbb{R}^3$ at each $p \in \mathbb{R}^3$.

The connection 1-forms of the frame field are:

hurry
Dad

$$(\omega_{ij})_p(v) := (\nabla_v E_i)_p \cdot (E_j)_p.$$

1-form   tangent vector at $p \in \mathbb{R}^3$

" ω_{ij} is the 1-form which, when evaluated on a tangent vector $V \in T_p\mathbb{R}^3$, tells you the j^{th} component of how E_i varies in the direction of V

$\rightsquigarrow$ it's a matrix of 1-forms, $\omega = \begin{pmatrix} \omega_{11} & \omega_{12} & \omega_{13} \\ \omega_{21} & \omega_{22} & \omega_{23} \\ \omega_{31} & \omega_{32} & \omega_{33} \end{pmatrix}$

$$\boxed{\text{Theorem} \quad \omega = dA\, A^T}$$

Exercise

$$E_1 = a_{11} U_1 + a_{12} U_2 + a_{13} U_3$$
$$E_2 = a_{21} U_1 + a_{22} U_2 + a_{23} U_3$$
$$\boxed{A_{ij} = E_i \cdot U_j}$$

$$\omega_{ij} = \sum_k \omega_{ij}(U_k)\, dx_k; \quad \omega_{ij}(U_k) = \nabla_{U_k} E_i \cdot E_j$$
$$= -E_j \cdot \nabla_{U_k} E_i$$
$$= -E_j \cdot \nabla_{U_k}\left[\sum_p a_{ip} U_p\right]$$
$$= -E_j \cdot \sum_p \left(\frac{\partial a_{ip}}{\partial x_k} U_p + 0 \right)$$

$$\therefore \quad \omega_{ij} = -\sum_{p,k} \frac{\partial a_{ip}}{\partial x_k} \underbrace{E_j \cdot U_p}_{a_{jp}}\, dx_k$$
$$= -\sum_p (dA)_{ip} A^T_{pj}$$

2.8. <u>The Structural Equations</u> Let E_1, E_2, E_3 be a frame field on $\mathbb{R}^3$, and $\theta_1, \theta_2, \theta_3$ its dual 1-forms, and ω_{ij} its connection 1-forms. Then

1. $d\theta_i = \sum_j \omega_{ij} \wedge d\theta_j$

2. $d\omega_{ij} = \sum_k \omega_{ik} \wedge \omega_{kj}$

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \quad dx = \begin{pmatrix} dx_1 \\ dx_2 \\ dx_3 \end{pmatrix}$$

$$\theta = A\, dx.$$

<u>Proof</u> (1) $d(dx) = 0$

$$\therefore \quad d\theta = d(A\, dx)$$

But, $U_j = \sum_k A_{kj} E_k$

$$\theta_i = \sum_j \theta_i(U_j)\, dx_j$$
$$= \sum_{j,k} A_{kj} \underbrace{\theta_i(E_k)}_{\delta_{ik}}\, dx_j$$

$$= \sum_j A_{ij}\, dx_j \qquad \therefore \quad \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix} = \underbrace{\begin{bmatrix} A_{11} & A_{12} \\ & \vdots \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} dx_1 \\ dx_1 \\ dx_c \end{bmatrix}}_{dx.}$$

$$\begin{aligned} \therefore \quad d\theta &= d(A\,dx) \\ &= dA \wedge dx + A \underbrace{d^2 x}_{=0} \\ &= dA\, A^T A \wedge dx \\ &= \underbrace{dA\, A^T}_{=\omega} \wedge \underbrace{A\,dx}_{\theta}. \end{aligned}$$

$$d(f\omega) = df \wedge \omega + f\, d\omega.$$

$$\begin{aligned} (2) \quad d\omega &= d(dA\, A^T) \\ &= -\,dA \wedge dA^T \\ &= -\,\omega A \wedge A^T \omega^T \\ &= -\,\omega \wedge \omega^T \\ &= \omega \wedge \omega. \end{aligned}$$

$$\omega = dA\, A^T$$

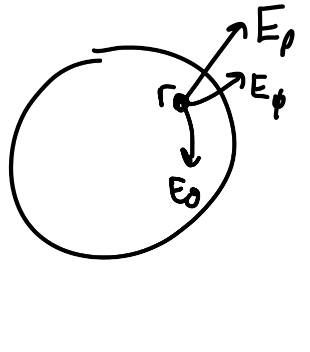
$$\boxed{\omega^T = A\, dA^T}$$

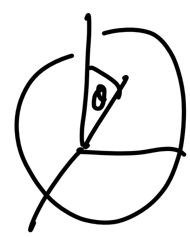
$$\therefore \; dA^T = A^T \omega^T$$

i.e. $dA = \omega A$

$$\begin{aligned} d(df.g) &= d(g\,df) \\ &= dg \wedge df \\ &= -\,df \wedge dg. \end{aligned}$$

eg. Spherical frame field





$\theta = A\,dx.$

$$\begin{aligned} x &= r \sin\theta \cos\phi \\ y &= r \sin\theta \sin\phi \\ z &= r \cos\theta \end{aligned} \quad \Rightarrow \quad \begin{aligned} E_r &= \text{normalize}\left(\frac{\partial p}{\partial \rho}\right) = \frac{xU_1 + yU_2 + zU_3}{r} \\ E_\theta &= \text{normalization of } \frac{\partial p}{\partial \theta} \end{aligned}$$

$A =$

If E_i is the frame field of a coordinate system $p = p($

$\theta = A\,dx.$

Claim $\theta_a = \sum_i \frac{\partial p}{\partial x_i}$ d

$$A = \begin{bmatrix} E_i \cdot U_j \\ \\ \end{bmatrix}$$

Lesson

Example

And here's some writing

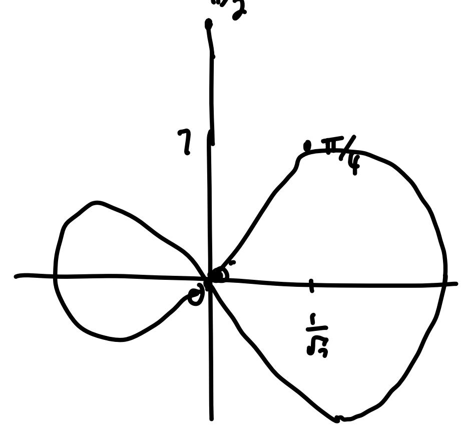
a.mp4

And some more.

4.1 Surfaces

Defn A coordinate patch $x : D \to \mathbb{R}^3$ is an injective immersion of an open set $D \subseteq \mathbb{R}^2$ into $\mathbb{R}^3$.

Defn A subset $M \subseteq \mathbb{R}^3$ is called a surface if for each $p \in M$ there exists



$(\sin t, \sin 2t)$

$\pi/4$:	$\frac{1}{\sqrt{2}}$	1
$\frac{\pi}{2}$	1	0
$\frac{3\pi}{4}$	$\frac{1}{\sqrt{2}}$	-1
π		